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PERFORMANCE ANALYSIS OF A TURBOJET ENGINE

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ABSTRACT

This experiment analyzes the performance of a small SR-30 turbojet engine operating on the Brayton cycle using sample data from the Turbine Technologies Mini-Lab system. The objective is to determine density, velocity, volumetric flow rate, mass flow rate, and Mach number at the compressor inlet and thrust nozzle exit, and to compare a momentum-based thrust prediction with thrust measured by the Mini-Lab load cell. Using pitot-static measurements, static pressures and temperatures, and known cross-sectional areas, we calculated inlet and exit flow properties for a single high-power operating point. A one-dimensional steady-flow control-volume model was then used to estimate theoretical thrust. The results show subsonic flow at both stations ($M_1 \approx 0.16$, $M_5 \approx 0.36$) and a theoretical thrust of about 9.5 lbf, slightly lower than the measured thrust of about 11 lbf, consistent with losses and uncertainties reported in previous SR-30 studies. This shows that the Brayton-cycle and gas-dynamics relations from class can be used to make reasonable predictions for the SR-30.

Keywords: Brayton cycle, SR-30 turbojet, Mini-Lab gas turbine, pitot-static measurement, mass flow rate, Mach number, thrust

Nomenclature

p_s – *Static pressure (Pa, kPa, psia)*
p_t – *Total (stagnation) pressure (Pa)*
T – *Static temperature (K, R)*
ρ – *Density (kg/m³)*
V – *Velocity (m/s, ft/s)*
$\dot{V}$ – *Volumetric flow rate (m³/s, ft³/s)*
$\dot{m}$ – *Mass flow rate (kg/s, lbm/s)*
A – *Cross-sectional area (m²)*
M – *Mach number (–)*
γ – *Ratio of specific heats (–)*
R – *Specific gas constant for air (J/(kg·K))*
F_{theory} – *Theoretical thrust (N, lbf)*
F_{meas} – *Measured thrust from load cell (N, lbf)*

1. INTRODUCTION

The objective of this experiment is to analyze the performance of the SR-30 turbojet at a single high-power operating point. Using Mini-Lab sample data, we determine density, velocity, volumetric flow rate, mass flow rate, and Mach number at the compressor inlet and nozzle exit, and we compare momentum-based thrust predictions with thrust measured by the Mini-Lab load cell. The SR-30 is a laboratory-scale turbojet integrated into the Turbine Technologies Mini-Lab™ system, which includes a test stand, instrumentation, data acquisition, and a thrust balance. The goal of this experiment was to connect the ideal Brayton-cycle concepts from class to the behavior of a real gas turbine by examining how air flows into the compressor and out of the exhaust nozzle, and how these flows relate to the engine's thrust and overall performance.

2. THEORY

The SR-30 turbojet operates on the Brayton cycle, which is the basic thermodynamic cycle for gas-turbine engines. In its simplest form, the ideal Brayton cycle consists of four processes [4,5]:

1. Isentropic compression (1→2): Ambient air is drawn into the compressor and compressed to a higher pressure and temperature.
2. Constant-pressure heat addition (2→3): Fuel is injected and burned in the combustor at approximately constant pressure, raising the gas temperature significantly.
3. Isentropic expansion in the turbine (3→4): The high-temperature gas expands through the turbine, producing shaft work to drive the compressor.
4. Isentropic expansion in the nozzle (4→5): The remaining energy in the gas is converted into kinetic energy in the exhaust nozzle, producing a high-speed jet and thrust.

If the cycle were perfectly ideal, all compression and expansion would be isentropic (no losses), pressure drops in the combustor would be negligible, and specific heats would be constant. Under these assumptions, the Brayton cycle efficiency can be expressed in terms of the compressor pressure ratio alone [4,5]. In a real engine like the SR-30, however, component efficiencies are less than 100%, there are pressure losses in the combustor and ducts, and heat is lost to the surroundings. These effects reduce the actual thermal efficiency and thrust compared to the ideal case.

In a turbojet, most of the turbine's work is used to drive the compressor; only the remaining energy appears as kinetic energy in the exhaust jet. The thrust generated can be approximated using a one-dimensional control-volume analysis as the difference in momentum flux between the exhaust and the inlet, plus a pressure term at the nozzle exit [4]. One of the main goals of this lab is to see how well that ideal thrust expression matches the thrust measured by the load cell under actual SR-30 operating conditions.

3. METHODS

3.1 Engine and Equipment

The engine used in this experiment was the Turbine Technologies SR-30 turbojet, installed in the Mini-Lab™ Gas Turbine Power System [1]. The SR-30 is a compact, single-spool turbojet commonly used in teaching laboratories [2,3]. Its primary components include a centrifugal compressor that draws in ambient air and raises its pressure, an annular combustor where fuel is mixed with the compressed air and burned at nearly constant pressure, a single-stage axial turbine that drives the compressor, and a converging exhaust nozzle that accelerates the flow to produce thrust.

The Mini-Lab test stand houses the engine, fuel and lubrication systems, and a thrust balance with a load cell for measuring static thrust. Instrumentation is installed at several standard stations along the flow path and typically includes thermocouples at the inlet, compressor exit, turbine inlet, turbine exit, and exhaust (T1–T5), static and total pressure taps at the compressor inlet and exhaust (P1–P5), a fuel-flow meter, and a tachometer for engine speed. All sensor outputs are routed to a data-acquisition system and recorded via a LabVIEW interface. For this project, live operation of the SR-30 was not available; instead, the manufacturer's Mini-Lab sample data file was used, which provides time histories of all relevant variables for a representative high-speed engine run [1–3]. In what follows, the subscripts 1 and 5 are used to denote the compressor inlet and the nozzle exit, respectively, consistent with the Mini-Lab notation.

3.2 Experimental Procedure and Data Processing

Under normal conditions, the experiment would begin with a standard SR-30 start-up and operating sequence. Pre-start checks of fuel, oil, and air-start systems would be completed, the appropriate LabVIEW VI loaded, and the engine started with instructor assistance. After stabilizing at idle, the throttle would be advanced through several power settings, including a high-rpm condition suitable for performance analysis, while LabVIEW records the time histories of pressures, temperatures, fuel flow, speed, and thrust.

Because this study relied on the Mini-Lab sample data rather than a live run, the experimental work focused on selecting and processing a representative operating point. The sample data file was opened, and the time histories were exported to Excel with separate columns for time, engine speed, pressures (P1–P5), temperatures (T1–T5), fuel flow, and thrust. Engine speed and thrust were plotted versus time to identify a period of high and steady rpm. A single time step within this plateau was chosen as the analysis point, and the corresponding inlet and nozzle-exit pressures and temperatures, fuel flow, and thrust were extracted.

Using the static pressures and temperatures at the compressor inlet and nozzle exit, air density at each station was calculated from the ideal-gas relation. The difference between total and static pressure measured by the pitot-static probes was then used to compute local velocity. From the velocity and known cross-sectional areas, volumetric flow rate, mass flow rate, and Mach number were obtained at both stations. Finally, a one-dimensional momentum balance was applied to estimate theoretical thrust from the inlet and exit flow properties, and this prediction was compared to the thrust measured by the Mini-Lab load cell at the same instant.

4. EXPERIMENTS RESULTS AND DISCUSSION

4.1 Results

The calculated compressor inlet and nozzle-exit properties are summarized in Table 1. At the inlet, the air density is about 1.3 kg/m³ and the velocity is only about 52 m/s, giving a Mach number of approximately 0.16. At the nozzle exit, the density drops to about 0.58 kg/m³ while the velocity increases to 189 m/s, corresponding to a Mach number of approximately 0.36. These values show a significant increase in velocity between the compressor inlet (station 1) and nozzle exit (station 5), while the Mach numbers in Table 1 confirm that the flow remains subsonic at both stations. These trends make sense: the large inlet area keeps the inlet Mach number well below 1, and the converging nozzle accelerates the flow in a way that matches published SR-30 data [2,3]. The mass flow rates calculated at the inlet and exit are similar in magnitude, confirming approximate conservation of mass and indicating that the sample data are internally consistent.

Table 1 Compressor inlet and nozzle-exit properties at the selected operating point

Quantity	Inlet (1)	Exit (5)	Units
p_s	103	112	kPa
T	263	678	K
ρ	1.3	0.58	kg/m³
V	52	189	m/s
$\dot{V}$	0.16	0.47	m³/s
$\dot{m}$	0.21	0.27	kg/s
M	0.16	0.36	–

4.2 Sample Calculations

To show how the main values were obtained, two calculations were performed.

4.2.1 Mach number at the compressor inlet

First, the speed of sound was calculated from the measured inlet temperature.

Given: $V_1 = 52$ m/s, $T_1 = 263$ K, $\gamma = 1.4$, $R = 287$ J/(kg·K)

Speed of sound:

$$a_1 = \sqrt{\gamma R T_1}$$
$$a_1 = \sqrt{(1.4)(287)(263)} \approx 325\ m/s$$

Mach number:

$$M_1 = \frac{V_1}{a_1} = \frac{52}{325} \approx 0.16$$

So, the inlet Mach number is approximately $M_1 \approx 0.16$ which agrees with the low-speed, subsonic inlet flow described in the results.

4.2.2 Theoretical thrust from momentum terms

Next, the thrust was estimated using the inlet and exit momentum terms from the control-volume momentum equation. For the chosen operating point, the momentum terms were:

Given:

$$\dot{m}_1 V_1 = 2.42\ lbf\ [Inlet], \quad \dot{m}_5 V_5 = 11.9\ lbf\ [Exit]$$

Ignoring small pressure-correction terms for this sample, the theoretical thrust is:

$$F_{theory} = \dot{m}_5 V_5 - \dot{m}_1 V_1$$

$$F_{theory} = [11.9 - 2.42][lbf] = 9.48\ lbf \approx 9.5\ lbf$$

Converting to SI units using 1 lbf = 4.448 N:

$$F_{theory} \approx 9.5\ lbf\ \times 4.448 \frac{N}{lbf} \approx 42.2N$$

This matches the theoretical thrust value reported in the results section.

4.3 Ideal vs. Actual Comparison

Table 2 compares theoretical thrust from the one-dimensional momentum equation to the thrust measured by the Mini-Lab load cell at the same time step. The theoretical thrust was found to be about $F_{\text{theory}} \approx 9.5$ lbf, while the measured thrust was approximately $F_{\text{meas}} \approx 11$ lbf.

Using the measured value as the reference, the percent difference is

$$\%\ diff = \frac{|F_{meas} - F_{theory}|}{F_{meas}} \times 100\% = \frac{|11 - 9.5|}{11} \times 100\% \approx 14\%$$

A difference of roughly 14% is reasonable for this type of simplified analysis. The theoretical thrust is based on a one-dimensional, steady, uniform flow model that uses pitot tube data at a single point in the exhaust. In contrast, the load cell measures the net force produced by the entire exhaust jet and includes the effects of flow non-uniformity, secondary flows, pressure errors at the nozzle exit, and mechanical losses in the engine and test stand. As a result, it is expected that the ideal thrust will be lower than the measured thrust.

Overall, the ideal and actual thrust values are nearly the same magnitude and follow the same trend, which indicated that the control-volume momentum model provides a reasonable estimate of SR-30 thrust while still showing the impact of real engine losses and measurement effects.

Table 2 Theoretical vs measured thrust at the selected operating point

Thrust quantity	Value (lbf)	Value (N)
Inlet momentum, $\dot{m}_1 V_1$	2.42	10.8
Outlet momentum, $\dot{m}_5 V_5$	11.9	51
Theoretical thrust, F_{theory}	9.48	42.2
Measured thrust, F_{meas}	11	49

4.4 Discussion

From the results in Table 1, it was clear that the SR-30 operated with subsonic flow at both the compressor inlet and the nozzle exit. The inlet velocity and Mach number were relatively low, which makes sense because the inlet area is large and is designed to bring air smoothly into the compressor without choking. At the nozzle exit, the flow speed and Mach number increased significantly, showing that the converging nozzle accelerates the exhaust and raises the dynamic pressure. However, even at this higher speed the Mach number stayed below 1, which is expected for a purely converging nozzle and agrees with basic compressible-flow theory.

The mass flow rates at the inlet and exit were similar, supporting the conservation of mass and suggesting that the sample data set is internally consistent. The thrust breakdown in Table 2 showed that the exit momentum term dominates the inlet term, which matches our physical intuition that most of the thrust in a turbojet comes from the high-speed exhaust jet. The theoretical thrust from the one-dimensional momentum equation was on the same order as the thrust measured by the load cell, indicating that the simple control-volume model captures the main behavior of the engine.

Some differences between the thrust calculated from pitot-tube readings and the thrust measured by the system's load cell are expected. The pitot tube only measures conditions at one position in the exhaust stream, not across the full cross-section of the nozzle. If the velocity and pressure are not perfectly uniform, the local dynamic pressure at the probe will not represent the area-averaged conditions. In addition, the probe can be slightly misaligned with the flow, and there are uncertainties in the pitot calibration, nozzle area, and the assumption of steady, one-dimensional flow. The load cell, on the other hand, responds to the total force produced by the exhaust on the engine and stand. Because it effectively "integrates" the momentum flux over the entire jet, it gives a different value than a thrust estimate based on a single pitot reading unless a correction factor is applied. This explains why the calculated thrust and the load-cell thrust do not match exactly.

The exhaust-gas temperature also affects the calculated Mach number. For a calorically perfect gas, the speed of sound is given by $a = \sqrt{\gamma RT}$, so temperature is the main variable that sets the local value of a. For a given exhaust velocity, if the temperature is higher, the speed of sound is higher and the Mach number $M = V/a$is lower; if the exhaust is cooler, the speed of sound is lower and the Mach number is higher. This means that any error in the measured exhaust temperature directly changes the Mach number, and small temperature variations across the jet can cause the "true" Mach number to differ from the value calculated using only a single thermocouple reading.

Overall, the results followed the expected trends. The nozzle produced a substantial increase in velocity and Mach number while remaining subsonic, the exit momentum dominated the thrust, and the differences between calculated and measured thrust could be explained by probe placement, flow non-uniformity, and measurement uncertainty, similar to how measurement effects influenced the ideal vs. actual cooling coefficients in the cross-flow heat-transfer lab.

4.5 Conclusion

This experiment used the Mini-Lab SR-30 sample data to connect Brayton-cycle and gas-dynamics concepts to the performance of a real turbojet engine. Even though the on-campus engine was not run during this session, the same data-reduction steps that would be applied to a live test were followed. Using pressures, temperatures, and the known geometry, we determined the density, velocity, volumetric flow rate, mass flow rate, and Mach number at the compressor inlet and nozzle exit. The results showed subsonic operation at both stations, with a clear increase in velocity and Mach number across the nozzle, which matches the expected behavior of a converging exhaust nozzle.

A steady-flow momentum balance was then used to estimate the theoretical thrust, which was compared to the thrust measured by the Mini-Lab load cell. The two values were of the same order of magnitude but did not agree exactly because of flow non-uniformity, probe placement, and other losses that are not captured in a one-dimensional model. Despite these differences, the overall trends were consistent with turbojet theory and with published SR-30 data. In summary, the lab reinforced how the Brayton cycle, compressible-flow relations, and momentum conservation work together in an actual gas-turbine engine and highlighted the importance of careful instrumentation and data interpretation when evaluating engine performance.

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